Luqman Nurhakim Bin Sairi

Mobile: +65 81218166 | Email: luqman.nurhakimbs@gmail.com
LinkedIn: LuqmanNurhakim

EDUCATION

Nanyang Technological University, Singapore Aug 2023 – July 2026
Bachelor of Engineering in Electrical and Electronics

- Core Modules: Data Structure & Algorithms, From Computational Thinking to Programming, Design & Innovation Project, Design & Innovation Project, Computer Communications
- Specialised Modules: Natural Language Processing, Web Application Design, Enterprise Network Design, Database Systems, Information Security

Nanyang Polytechnic, Singapore (Specialisation: Software Engineering Practices) Apr 2017 – May 2020
Diploma in Multimedia and Info Communication Technology

- Core Modules: Java Programming, Database Design, UX Design for Web Development, Web Development, Data Structure & Algorithms, Full Stack Application Development
- Specialised Modules: Cross-Platform Mobile App Development, Web API Development, Database Modelling & Implementation, Software Security & Testing

EXPERIENCE

Inland Revenue Authority of Singapore May 2025 – Jul 2025
System Integration and Test Engineer

- Developed and executed 50+ end-to-end integration test cases across a multi-tier enterprise system, ensuring zero regression prior to deployment.
- Authored SQL queries to validate, patch, and manipulate backend relational databases, supporting data integrity across 3 staging and QA cycles.
- Audited deployment pipelines to track over 20 unreleased code changes, streamlining release governance and improving technical documentation.
- Documented testing outcomes for 5 program modules, delivering concise weekly progress reports to supervisory teams to maintain project alignment.

Singapore Armed Forces Aug 2021 – Aug 2023
Armskote IC for 8SIR

- Managed daily armoury operations over a 1.5-year tenure, maintaining 100% continuous accountability for a large inventory of highly sensitive operational assets and controlled equipment.
- Conducted meticulous equipment inspections and discrepancy reporting, consistently achieving zero-fault clearances during both scheduled quarterly and unannounced surprise audits.
- Modernized the asset management workflow by conceptualizing and implementing a digital tracking system, replacing manual ledgers to significantly reduce issuance time and eliminate logging errors.
- Enforced strict facility security protocols and preventative maintenance standards, mitigating operational risks and ensuring all controlled equipment remained in a state of high readiness.

National Computer Systems Pte Ltd Mar 2019 – Aug 2019
Software Tester and Helpdesk Attendant

- Collaborated with Enterprise Singapore and SAC to validate 15+ public sector digital web forms, ensuring secure and accurate processing of corporate UEN credentials.
- Managed the incident resolution lifecycle on Redmine for over 120 bug reports, actively triaging issues, isolating front-end and back-end defects, and verifying critical post-deployment hotfixes.
- Automated 40+ scenario-based test suites to streamline testing workflows, accelerating User Acceptance Testing (UAT) cycles by an estimated 20% and significantly reducing manual testing overhead.

PROJECTS

FYP: PhytoLens – An AI assisted mobile application to identify plants/ animals using the camera. Aug 2025 – Apr 2026

- Engineered "PhytoLens", an Android mobile application for real-time plant and insect identification, independently managing the full software development lifecycle over a two-semester academic period.
- Integrated the Gemini API to process user-submitted images and dynamic prompts, leveraging generative AI to deliver accurate, contextual biological analyses directly within the app.
- Developed the application's core architecture using Java in Android Studio, integrating mobile hardware capabilities like camera functions with backend processing.
- Executed over 20 structured test cases to rigorously validate in-app functions, resolving edge cases and ensuring a highly stable, bug-free user experience prior to final project submission.

Design and Innovation Project: Creating and AI-Assisted WeiQi/GO App Feb 2025 – Apr 2025
Programmer

- Collaborated within an 8-member team to develop an AI-assisted WeiQi (Go) educational web application designed to accelerate the learning curve for novice players.
- Engineered responsive and intuitive front-end interfaces utilizing HTML, CSS, and JavaScript, prioritizing a seamless and highly accessible user experience for beginners.
- Programmed core backend components and API routing using Python and Flask, facilitating the integration of an AI coaching engine to deliver real-time strategic feedback during gameplay.

COMPETITIONS

Tinkering Projects by Garage@EEE Jan 2026 – May 2026
Lead Developer

- Collaborated within a 4-member team to build an offline, AI-powered mock interview application designed to simulate personalized interview scenarios.
- Engineered the core backend using Python and Flask, integrating the open-source SadTalker API to generate realistic, audio-driven facial animations and lip-syncing from single, user-provided static images.
- Optimized local GPU processing pipelines to compute complex 3D motion coefficients and render high-fidelity video, ensuring fluid image articulation while maintaining strict data privacy through a completely offline architecture.

NTU CAO x Garena Hackathon 2026 Jan 2026 – Mar 2026
Programmer and Asset Designer

- Collaborated within a 4-member team to design and develop an immersive, interactive RPG on the Roblox platform.
- Secured industry recognition from Garena professionals, who evaluated the project and acknowledged its strong commercial and gameplay potential.
- Implemented custom character movement animations and core mechanics using Luau, significantly enhancing visual fluidity and the overall player experience.

Micro Mouse Competition (Techathon) Dec 2024 – Jan 2025
Lead Developer for the algorithm

- Engineered a fully autonomous, 14x14cm MicroMouse robot from scratch within a 4-member team, seamlessly integrating custom circuit boards and Arduino hardware components for competitive maze-solving.
- Spearheaded the core algorithmic development using C++ and Arduino, taking ownership of the robot's autonomous navigation and real-time decision-making logic.
- Implemented advanced pathfinding routines utilizing Dijkstra’s algorithm, successfully calculating the shortest and fastest optimal routes to autonomously navigate and solve complex maze configurations.

CO-CURRICULAR ACTIVITIES

NTU Garage@EEE Main-Committee for IdeasJam 2025 (Logistics) Dec 2024 – Apr 2025

- Managed end-to-end catering operations and physical table setups, ensuring seamless hospitality logistics and vendor coordination for the event.
- Designed strategic spatial layouts for food stations and guest seating, optimizing crowd flow and accessibility to enhance the overall event experience.
- Fabricated custom electronic Tokens of Appreciation, utilizing precision soldering and hardware assembly techniques to deliver unique, high-quality commemorative gifts.

NTU EEE Club Efinity Sub-Committee (Logistics) Mar 2024 - Aug 2025

- Coordinated the comprehensive distribution of event merchandise and participant apparel, ensuring accurate allocation and a seamless registration experience for all attendees.
- Managed on-site food and beverage logistics, proactively replenishing refreshments and provisions to maintain optimal guest satisfaction and comfort throughout the duration of the event.
- Collaborated with cross-functional committee portfolios, including the technical programming team, to synchronize event sequencing and ensure the seamless execution of interactive station games.
- Facilitated operational alignment between logistics and technical personnel, providing strategic support to optimize the overall flow of events and maintain high participant engagement.

SKILLS

- **Languages**: Python, Java, C++, C, SQL, JavaScript, HTML/CSS, Lua
- **Frameworks & Databases**: Flask, Node.js, Django, Android SDK, Firebase, MySQL, MongoDB
- **Hardware & Tools**: Arduino, Embedded C, PCB Soldering, 3D Printing, Git, Linux